

Saad Ahmed Qadeer

Data Scientist | AI Engineer

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Meta Hacker Cup Qualifier | Millennium Scholarship Holder | 10+ Global AI Hackathons

KEY SKILLS AND TECHNOLOGIES

Coding Languages:	Python, SQL, C++, Matlab, Latex
ML / AI:	PyTorch, TensorFlow, Scikit-Learn, XGBoost, HuggingFace, LangChain
Libraries:	Pandas, NumPy, SciPy, Matplotlib, Seaborn, Power BI
Dev & Tools:	Git, Docker, Linux, Jupyter, VS Code, AWS
Interests:	Machine Learning, Deep Learning, NLP, Generative AI (RAG), Computer Vision

EDUCATION

Bachelor of Science in Data Science	February 2021 – January 2025
<i>The Islamia University of Bahawalpur, Pakistan</i>	<i>CGPA: 3.5 / 4.0, Bronze Medalist</i>
• Undergraduate Thesis: “Automated Thoracic Disease Detection via Multilabel Deep Learning Ensembles”, Achieved 0.780 AUC using a Stacking Ensemble with Grad-CAM interpretability.	

WORK EXPERIENCE

Statistical Data Analyst	February 2025 – October 2025
<i>N5 Sensors</i>	<i>remote, USA</i>
• Collaborated with PhD researchers to optimize detection algorithm thresholds.	
• Maximized true-positive rates and suppressed false alarms in early fire warning systems.	
• Developed anomaly-detection pipelines and conducted LTE connectivity analysis (RSRP, RSRQ).	
• Performed end-to-end statistical analysis of real-world fire events and gas detection data.	
Independent Data Science Consultant	July 2022 – Present
<i>Upwork (International Clients)</i>	<i>Remote</i>
• Maintained a 100% Job Success Score across 6+ projects in predictive modelling and NLP.	

TEACHING EXPERIENCE

Instructor – “Advanced Statistics with Projects”	February 2026 – March 2026
<i>iCodeGuru, California, USA</i>	
• Design and taught an “Advanced Statistics with Projects” course to 40+ underprivileged students.	
• Guided students through practical data science applications and hands-on statistical modeling.	
• Strengthened public speaking and technical communication through daily presentations.	
• All course sessions are fully documented and recorded, accessible via Sheets .	
Instructor – “Introduction to Generative AI with LangChain and Hugging Face”	May 2026
<i>iCodeGuru, California, USA</i>	
• Delivered workshop title “Introduction to Generative AI with LangChain and Hugging Face”.	
• Covered concepts of generative AI, LangChain, and Hugging Face through practical demonstrations.	
• Recorded sessions are available at Workshop Recording .	

RESEARCH & ACADEMIC PROJECTS

Mechanistic Interpretability & In-Context Learning	<i>GPT-2-Small · TensorFlow · Python</i>
• Replicated Wang et al. (EMNLP '23) on GPT-2-Small, achieving 80.5% probing accuracy	
• Formalized the Label Surface Form Sensitivity (LSFS) metric to expose gaps in Anchor Theory.	
• Revealed an 11–12% accuracy collapse across semantically equivalent labels.	
• Engineered Canonical Anchor Projection , a training-free embedding correction.	
• Recovered 80–100% of the accuracy gap while maintaining the canonical baseline.	

Multilabel Chest X-Ray Pathology Classification 🤖

TensorFlow · Keras · DenseNet-121

- Built an end-to-end multilabel pipeline for 7,376 NIH ChestX-ray14 radiographs.
- Fine-tuned a DenseNet-121 and ResNet50V2 stacking ensemble using Asymmetric Loss.
- Achieved a **0.780 macro AUC-ROC (0.862 for Effusion)** for class imbalance.
- Validated model attention maps using GradCAM to maximize independent F1 scores.

De Novo RNA 3D Structure Prediction (Stanford AI Research) 🤖

PyTorch · RhoFold+

- Deployed a RhoFold+ pipeline, utilizing MSA subsampling to generate diverse structural ensembles.
- Designed a physics-informed random walk fallback for ultra-long sequences (>440 nt).
- Bypassed embedding limits and enforced valid C1' backbone geometry.
- Achieved a peak TM-score of **0.507** (correct global fold topology) on the 9IWF target.
- Set a **0.139 zero-shot baseline** without task-specific fine-tuning.

Graph-RAG: Mitigating LLM Hallucinations via Knowledge Graphs 🤖

Python · Neo4j

- Built a hybrid retrieval pipeline fusing Neo4j Knowledge Graphs with BGE-Large dense vector search.
- Improved **RAGAS Faithfulness to 0.871** (+17.4% vs. Naive RAG) on a multi-hop benchmark.
- Achieved a **+23.0% gain on 3-hop queries**, validated via McNemar's test and bootstrap intervals.
- Deployed a containerized, full-stack system exposing live A/B testing REST APIs.

Probabilistic RUL Estimation for Turbofan Engines 🤖

PyTorch · MLflow · Numpy

- Engineered a Bayesian LSTM with MC Dropout to approximate deep Gaussian posteriors.
- Achieved a **competitive 15.18 RMSE** on the NASA C-MAPSS FD001 benchmark dataset.
- Prevented data leakage via train-only scaling and sliding-window tensor construction.
- Quantified epistemic uncertainty with 95% prediction intervals that narrow near failure.
- Bounded catastrophic late-prediction risks utilizing the **NASA Asymmetric Score**.

HONORS & INTERNATIONAL COMPETITIONS

- Earned a **Bronze Medal** for outstanding academic performance in undergraduate studies.
- Ranked **42nd globally** out of 13,779 participants in the **Meta Hacker Cup 2025 (AI Track)**
- **Harvard University CS50x** Puzzle Day Achiever
- Completed **Millennium Fellowship**, a United Nations Academic Impact leadership program.
- Participated in **10+ international competitions** including Stanford, Google DeepMind, Kaggle, etc.
- Achieved **Top-Rated** status as a Freelance Data Science Consultant, securing a 100% Job Success Score.

HOBBIES

- Morning walks, cooking, and recreational sports (cricket and badminton).